Introduction to Statistical Investigations

Second Edition

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Chapter 2.2, 2.3

Generalization: How Broadly Do the Results Apply?

Quantitative Data / Inference for a Single Quantitative Variable

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Warm Up

- Let's say you want to find out how many hours SMC students sleep. You want to find out if the average number of hours all students sleep is less than 8.

A. How would you obtain a sample?

B. What variable would you measure on each student? Is it quantitative, or categorical?

C. Give an example of *convenience* sampling in this scenario.

D. Give an example of *simple random* sampling in this scenario.

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Example 2.2

Sampling Students

- Last time we looked at the categorical variable “whether or not the student lives on campus” for all students living in a small midwestern college.
- We will now look at the quantitative variable “cumulative GPA” for all students enrolled that particular semester.

✅/❎ vs. 📈

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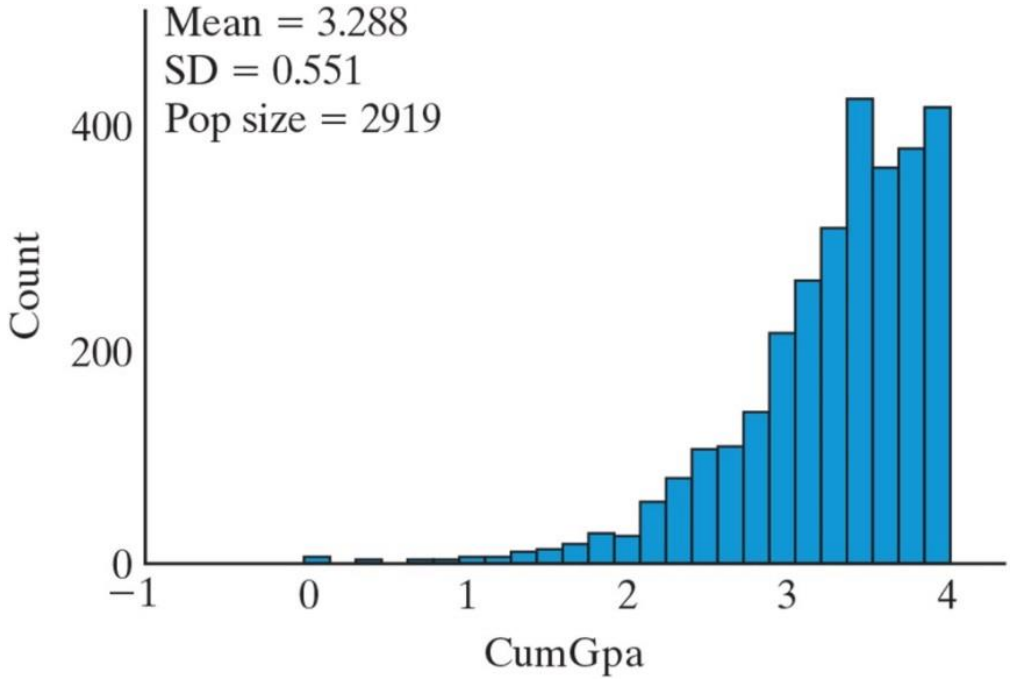
Example 2.2: Sampling Students (1 of 2)

- The registrar gave us all 2,919 students' cumulative GPAs.
- Here's the distribution:



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Example 2.2: Skewed

- A distribution of data is **skewed** if it is not symmetric
  - Bulk of the values tend to fall on one side of the distribution with the "longer tail"

- **Right-skewed** distributions have their longer tail *on the right*.
- **Left-skewed** distributions have their longer tail *on the left*.
- The distribution of cumulative GPAs is highly skewed to the left.

Skew = the long tail

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Example 2.2: Median

- Sort the data from smallest to largest → Grab the **middle** value
- Middle value is known as the **median**.

- The median of the GPA data is 3.41 (the 1460th observation) in the sorted GPAs.
- So half the students have a GPA of 3.41 or higher.
- The mean of the distribution is lower at 3.288.
- The mean gets pulled in the direction of the skewness compared to the median.

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Example 2.2: Standard Deviation

- Remember that often we think of standard deviation as the "typical" distance from the mean.
- But we should be cautious in doing this with a skewed distribution like we have.
- The mean isn't where it is on the normal distribution!

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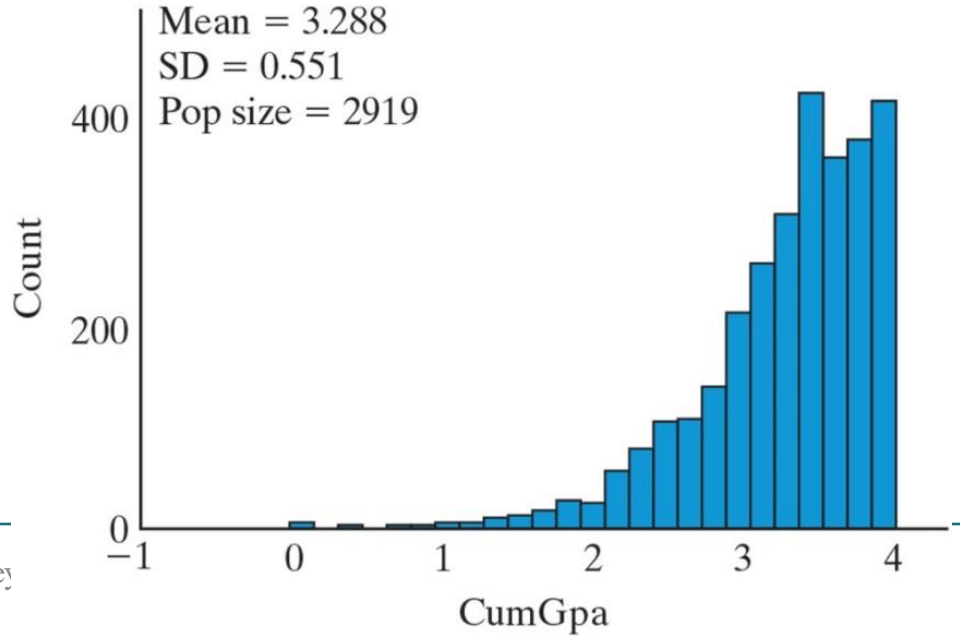
Example 2.2: Symbols

- Population mean: μ ("mu")
  - Mean is just "average" – easy to compute wherever we are.
- Population standard deviation: σ ("sigma")
- The population mean GPA is μ =3.288
- The population standard deviation GPA is σ = 0.551.
  - These numbers come from the applet



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Example 2.2: Sample of 30 Students (1 of 2)

Let's pretend we didn't have access to the population, but only this random sample of 30 student GPAs.

ID	Cum GPA	ID	Cum GPA	ID	Cum GPA
827	3.44	844	3.59	825	3.94
1355	2.15	90	3.30	2339	3.07
1455	3.08	1611	3.08	2064	3.48
2391	2.91	2550	3.41	2604	3.10
575	3.94	2632	2.61	2147	2.84
2049	3.64	2325	3.36	2590	3.39
895	2.29	2563	3.02	1718	3.01
1732	3.17	1819	3.55	168	3.04
2790	2.88	968	3.86	1777	3.83
2237	3.25	566	3.60	2077	3.46

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Example 2.2: Symbols for Samples

- Sample mean: $\bar{x}$ ("x-bar")
- Sample standard deviation: s ("s")

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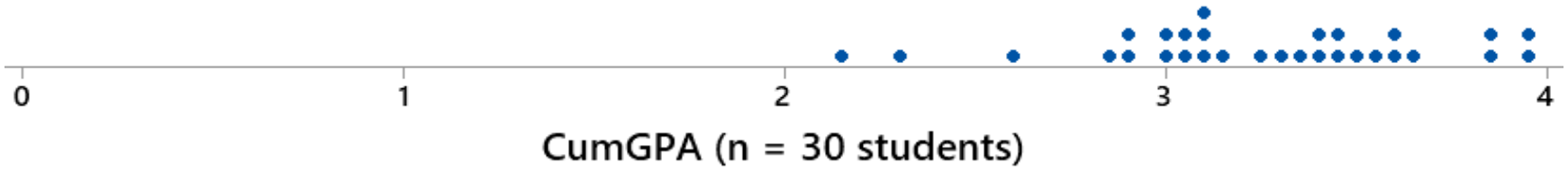
Example 2.2: Sample of 30 Students (2 of 2)

- Because we have a random sample, our sample shares some characteristics of the population. (which we know because the registrar gave us the graph)
- The sample distribution is also skewed left (see graph below).
- The sample mean is $\bar{x} = 3.243$ and the sample SD is $s = 0.438$.

We'll worry about how we got the SD later.



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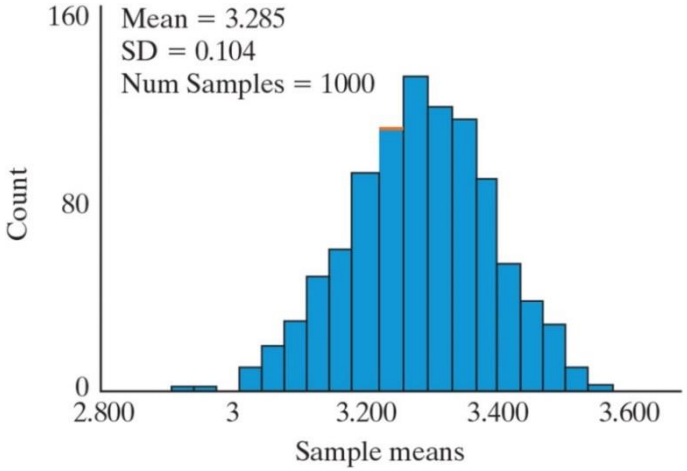
Example 2.2: Sampling Students (2 of 2)

- We know if we took a different sample of 30 students we would probably get a different mean.
- We want to know how these sample means might differ
- We took 1,000 random samples of 30 students and the graph shows the 1,000 mean GPAs from these samples.



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Example 2.2: Central Limit Theorem (1 of 2)

- The shape of the distribution of sample means is approximately normal even though the population was quite skewed.
- The ***Central Limit Theorem for sample means*** implies that the distribution of sample means is approximately normal when the sample size is large.
- It appears as though a sample size of 30 is large enough.
- If the population distribution is not highly skewed, the distribution of sample means will be approximately normal *if the sample size is at least 20.*

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Example 2.2: Mean of the GPAs

- The mean of our 1,000 GPAs was 3.285.
- This is quite similar to the population mean of 3.288.
- If we had an actual sampling distribution, instead of just 1,000 samples, these means would match exactly.

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Example 2.2: Standard Deviation of GPAs

- The standard deviation of our 1,000 samples was 0.104.
- Much smaller than the population SD of 0.551.
- This too, can be predicted. The SD of the sample means is predicted to be

$$\frac{\sigma}{\sqrt{n}} \text{ or } \frac{0.551}{\sqrt{30}} = 0.101 \text{ in our case.}$$

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Example 2.2: Central Limit Theorem (2 of 2)

- When taking random samples from a large population (more than 20 times the size of the sample), the distribution of sample means will:
  - Center around the population mean (μ)
  - Have a standard deviation smaller than the population, $\mathrm{SD}(\bar{x}) = \dfrac{\sigma}{\sqrt{n}}$
- **Central Limit Theorem:** The distribution of sample means will be approximately normal when the sample size is large (or if the population distribution is normally distributed)

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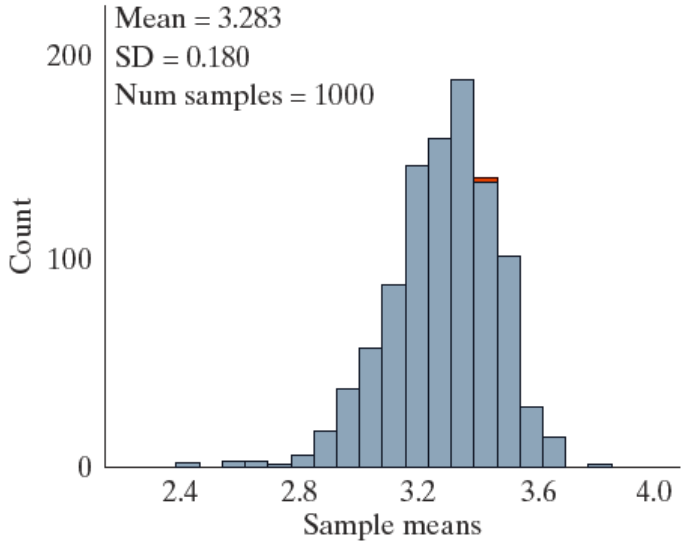
Example 2.2: What if the sample size was small?

- We took 1,000 samples of size 10 from the population.
- We do not get a symmetric distribution of sample means.
- This distribution is slightly skewed to the left



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Section 2.2: Summary

- With quantitative variables, sometimes the distribution is ***skewed***, and has a tail.
  - Skewed left = the tail is on the left
  - Skewed right = the tail is on the right.
- If we sort all the values we obtained, we can choose the middle value to get the ***median***.
- If we sample, we can computer the population mean $\bar{x}$ and the standard deviation.
- If our sample is large enough, >20, we can assume the variable will be approximately normal, then use the Central Limit Theorem.

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Section 2.3

Inference for a Single Quantitative Variable

- We will explore how to use the information from the previous section to carry out a theory-based test for a population mean.

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Example 2.3: Estimating Elapsed Time

- How long do you think a song is?
- Students in a stats class (for their final project) collected data on students' perception of time
- Subjects were told that they'd listen to music and then be asked questions when it was over.
- The researchers played 10 seconds of the Jackson 5's "ABC" and asked how long they thought it lasted
- Can students accurately estimate the length?

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Example 2.3: Hypotheses

Null Hypothesis: People will accurately estimate the length of a 10 second-song snippet, on average.
$(\mu = 10 \text{ seconds})$

Alternative Hypothesis: People will not accurately estimate the length of a 10 second-song snippet, on average.
$(\mu \neq 10 \text{ seconds})$

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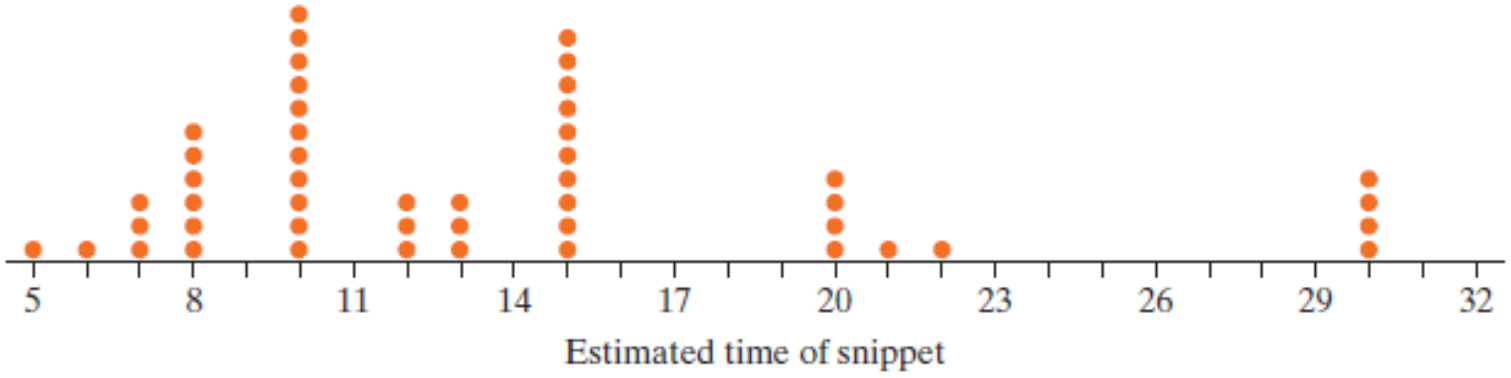
Example 2.3: Displaying Data

- A sample was recorded of 48 students on campus, each answering how long they estimated the song length to be.
- What does a single dot represent?
- What are the observational units? Variable?



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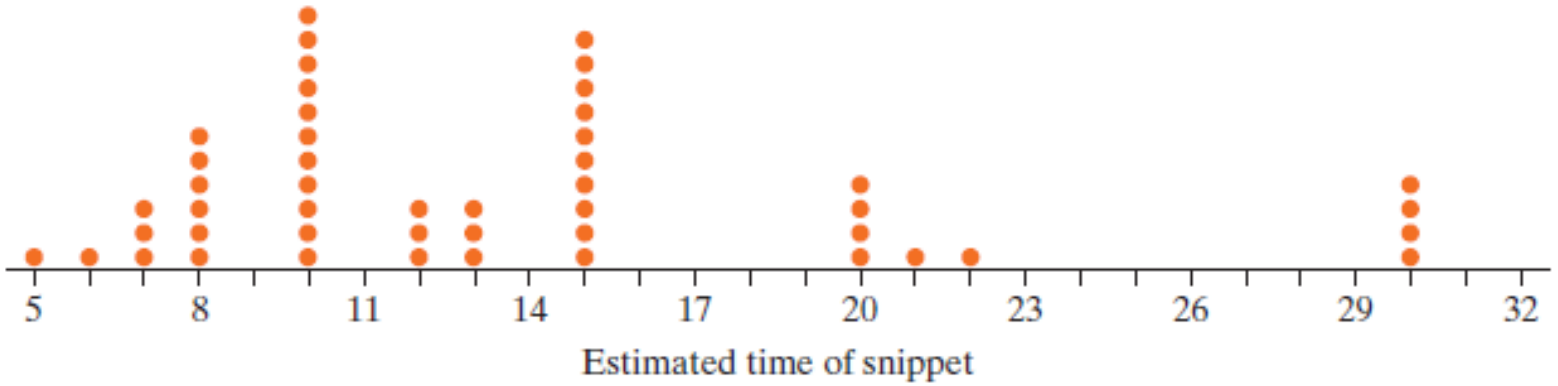
Example 2.3: Skewed, mean, median

- The distribution obtained is not symmetric but is **right skewed.**
- Remember that when data are skewed right, the **mean** gets pulled out to the right while the **median** is more resistant to this.
- Also notice there are spikes at 10 and 15 seconds.



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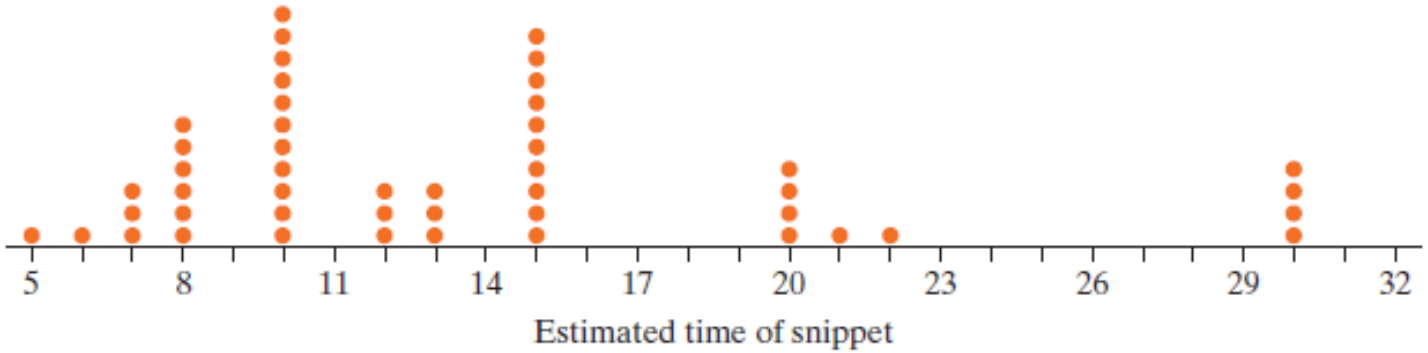
Example 2.3: Mean vs Median

- The mean is 13.71 and the median is 12.
- How would these numbers change if one of the people that gave an answer of 30 seconds actually said 300 seconds?
- The standard deviation is 6.5 sec. Is it resistant to outliers?



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Example 2.3: Inference (1 of 3)

- Can we generalize the results of this study to all college students in the population?
- Would all college students, on average, overestimate the actual length of the snippet (10 seconds)?
  - Was the sample randomly drawn from a population of all college students? No.
  - Is there an obvious reason to believe that these 48 students would estimate the length that much differently than the rest of the population? Probably not.
- So we will proceed as though we do have a representative sample.

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Example 2.3: Inference (2 of 3)

- $\mathbf{H_0} : \mu = 10$ seconds
- $\mathbf{H_a} : \mu \neq 10$ seconds
- Our problem now is, how do we develop a null distribution? (The second S in our 3S strategy.)
- Flipping coins and spinning spinners will not work to model what would happen under a true null hypothesis.
- Let's look at two ways to do this (the first way is really just a way to think about how this can be done).

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Example 2.3: Inference (3 of 3)

- We don't have population data.
- More specifically, we don't have population data that reflects our null hypothesis where $\mu = 10$ seconds.
- All we have is our sample of 48.

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Example 2.3: Simulation-Based Inference (1 of 6)

- We need to create a population based on what we know about our sample and what we want the mean to be.
- We need to come up with a large data set that we think our population of time estimates might look like **under a true null.**
- We might assume the population is skewed (like our sample) and has a standard deviation similar to what we found in our sample but has a mean of 10 seconds.

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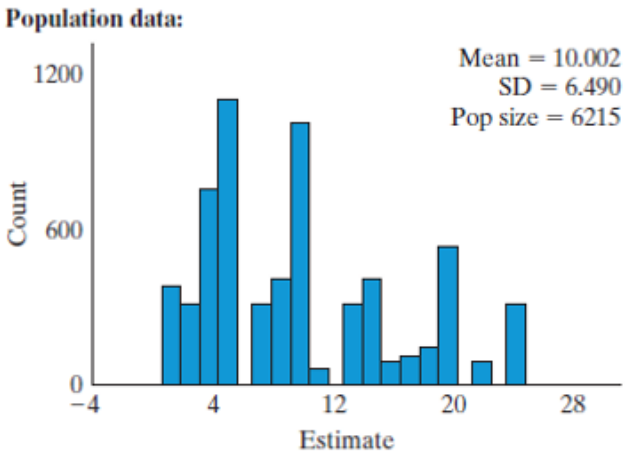
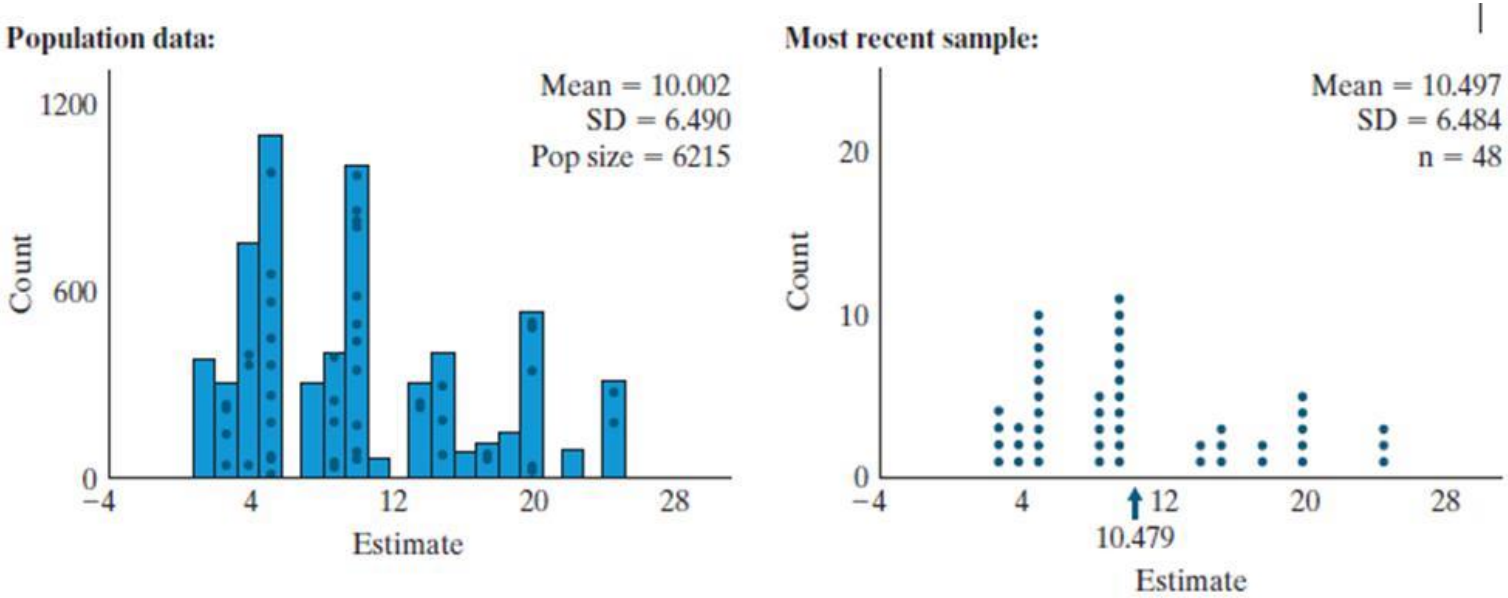
Example 2.3: Simulation-Based Inference (2 of 6)

- Invent a population of values with a mean of 10.
- Repeatedly, take random samples from that population to simulate what samples could look like when the null hypothesis is true.
- Here's simulated population data like what we need.
- It has a similar shape and SD to that of the sample.
- Notice, however, the mean of this distribution matches the value in the null hypothesis. (Which is what we want to develop a null distribution.)



Example 2.3: Simulation-Based Inference (3 of 6)

- To develop a distribution of statistics (sample means) that could have happened if the null was true, we start to take samples of size 48 from our made-up population.



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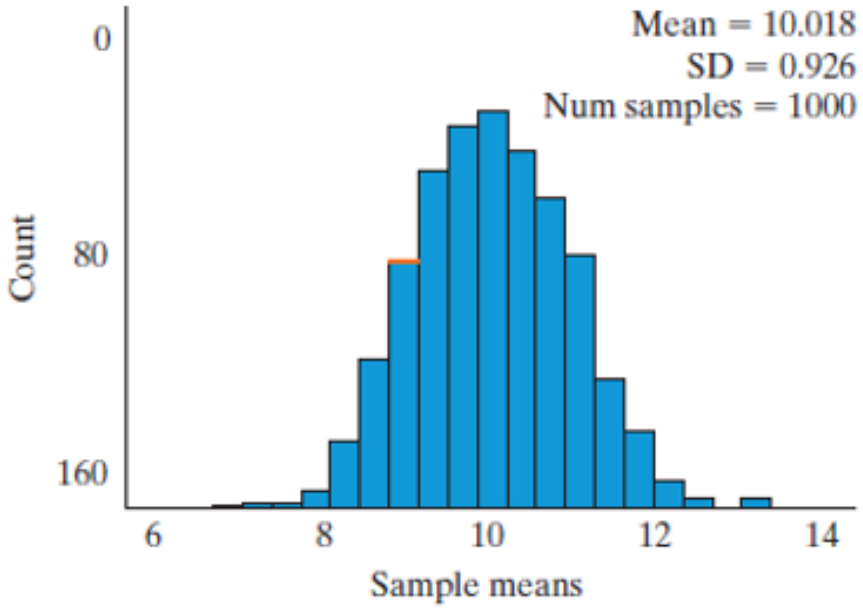
Example 2.3: Simulation-Based Inference

(4 of 6)

- Then repeat this so we develop a distribution of 1,000 sample means that could have happened if the null was true.



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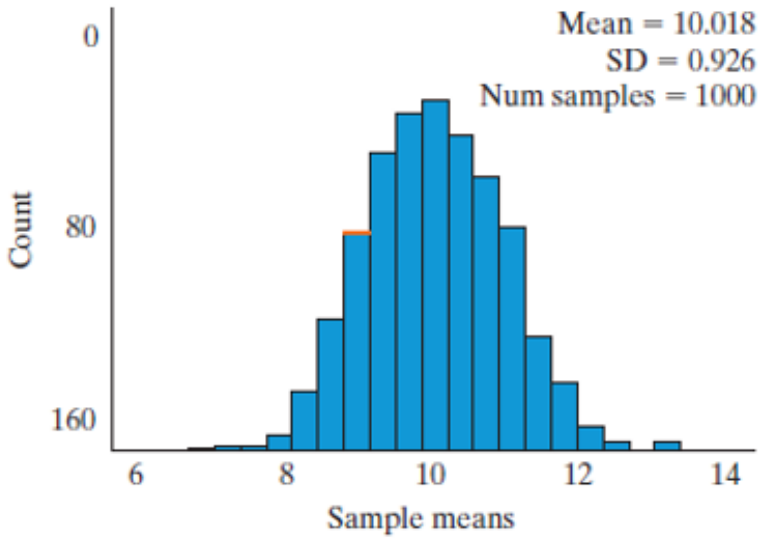
Example 2.3: Simulation-Based Inference (5 of 6)

- Notice that this distribution of sample means is bell-shaped, centered on the value of the null, has a standard deviation much smaller than that of the population standard deviation.
- Also notice that the sample mean of 13.71 seconds never occurred, so we have a *p*-value of about 0.
- What's the standardized statistic?
- What's our conclusion?



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Example 2.3: Simulation-Based Inference (6 of 6)

- Doing this simulation-based test for a single mean requires that we make a guess as to what the population distribution looks like.
- We did that to develop a distribution of simulated statistics.
- But can we predict what this distribution of simulated statistics is going to look like without developing a population distribution?
- Do we know what shape it will have? What its mean should be? What its standard deviation should be?

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Example 2.3: Theory-Based Test (1 of 3)

- The answer is yes to all these.
- In fact, creating a population each time we want to run a test of significance is extremely time consuming and cumbersome. (So this will be the only time we will do it.)
- In practice we will test a single mean using a theory-based test.

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Example 2.3: Theory-Based Test (2 of 3)

- As long as the sample size is large enough, we will see that we can predict the shape, center, and variability of a distribution of sample statistics given we just have a sample and not a population.
- The distribution of sample statistics:
  - Should be centered at the value under the null
  - If the sample size is large enough, it will be bell-shaped.
- We have seen that the standard deviation should be $\frac{\sigma}{\sqrt{n}}$ or $\frac{\sigma}{\sqrt{48}}$ in our case.

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Example 2.3: Theory-Based Test (3 of 3)

- Since we don't know the population standard deviation, our best guess as to its value is the sample standard deviation.
- So we will predict our distribution of statistics that could have happened if the null was true will have a standard deviation of $\frac{s}{\sqrt{n}}$ (s is the sample standard deviation)
- Because of this extra unknown (the population standard deviation) we will *not* use a normal distribution to approximate the distribution of sample statistics but will use a t-distribution.

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Example 2.3: *t*-distribution

- The shape of a *t*-distribution is very similar to a normal distribution.
- The *t*-distribution has slightly "heavier" tails than a normal distribution.
- The *t*-statistic is the standardized statistic we use with a single quantitative variable and can be found using the formula:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}}$$

The $\frac{s}{\sqrt{n}}$ (standard deviation of our sample divided by the square root of the sample size) is called the standard error and is an estimate for the standard deviation of the null distribution. (Notice that this formula has the same form as the one we used to determine the standardized statistic for a single proportion.)

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Example 2.3: *t*-statistic

- The t-statistic for our example is:

$$t = \frac{\bar{x} - \mu}{\frac{s}{\sqrt{n}}} = \frac{13.71 - 10}{\frac{6.50}{\sqrt{48}}} = \frac{3.71}{0.938} = 3.95$$

- With a standardized statistic of 3.95, we have strong evidence against the null.
- We can still use the rule that a standardized statistic more than 2 or less than negative 2 provides strong evidence against the null.

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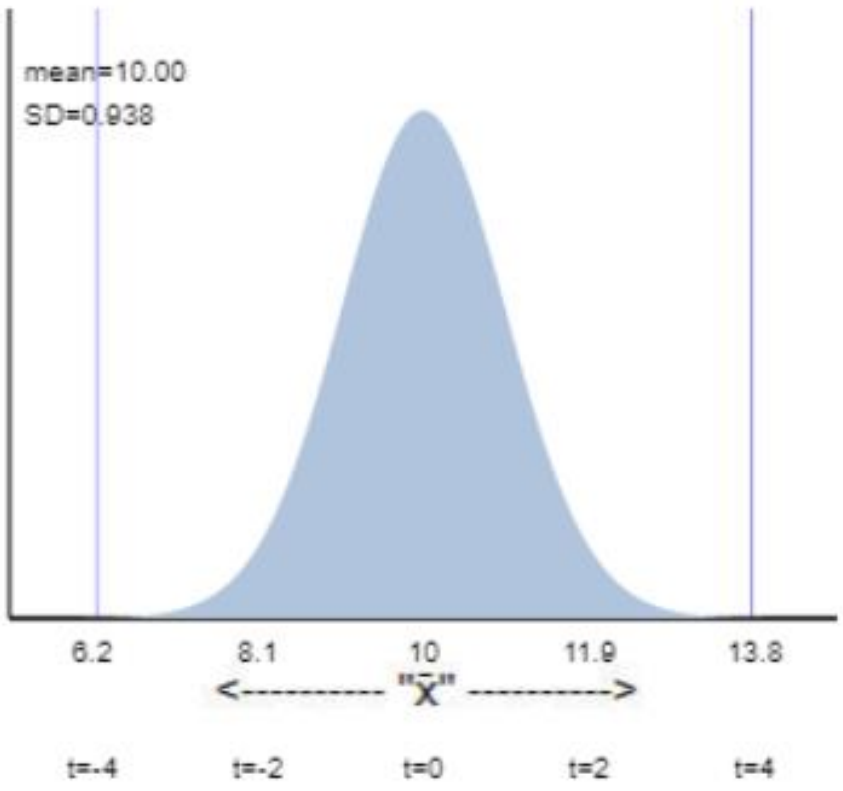
Example 2.3: *t*-distribution and *t*-statistic

- Here is a theoretical t-distribution for our example.
- Notice that our statistic of 13.71 seconds is way out in the tail of the distribution.
- Also notice the mean and SD of this distribution. How were these determined?



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Example 2.3: Validity Conditions

- The theory-based test for a single mean requires: **That you have a symmetric sample distribution, or a sample size of at least 20 and the sample distribution is not strongly skewed.**
- In practice we can use the theory-based applet to run this test. We can either enter the raw data or the sample size, sample mean, and sample standard deviation.
- (See results on the following slide.)

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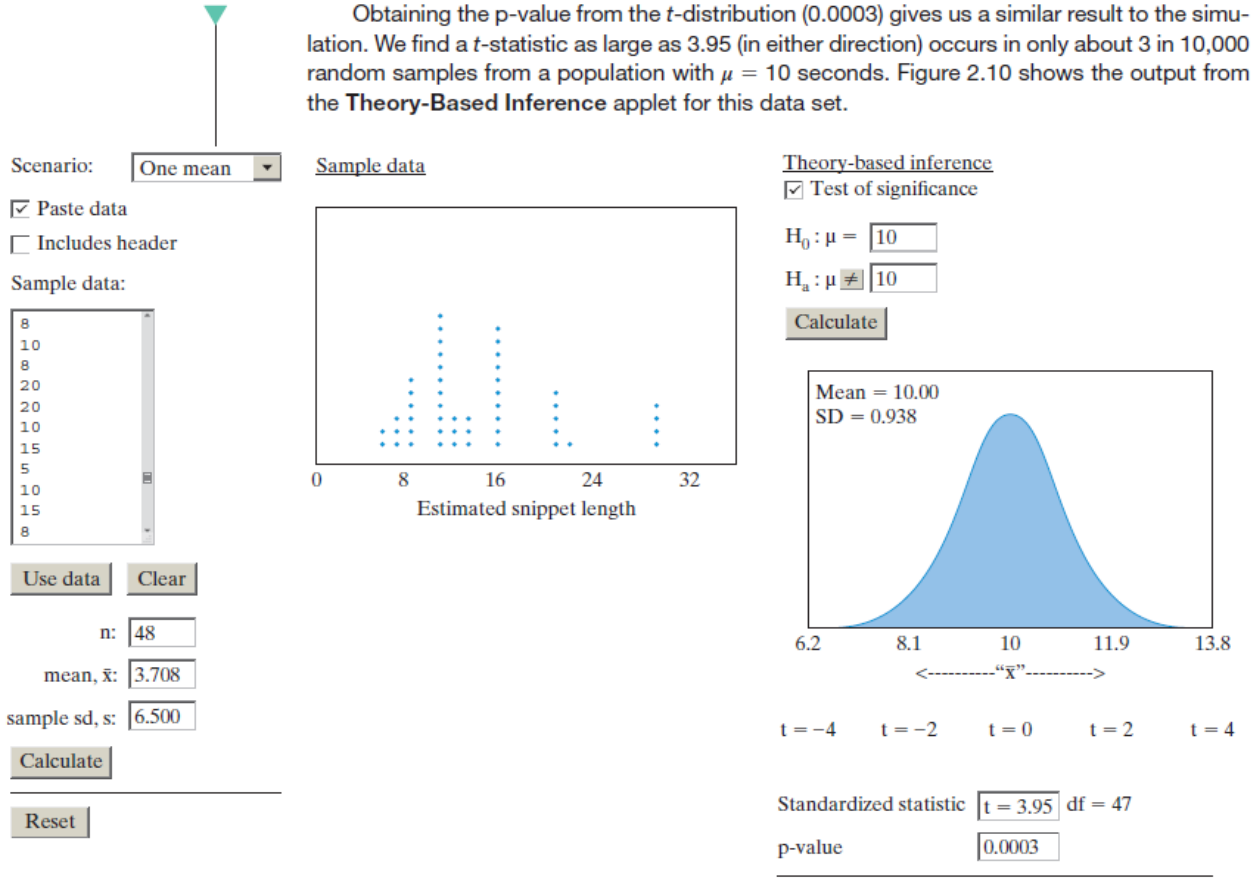
Example 2.3: Theory-based Applet: One Mean Scenario

Obtaining the p-value from the *t*-distribution (0.0003) gives us a similar result to the simulation. We find a *t*-statistic as large as 3.95 (in either direction) occurs in only about 3 in 10,000 random samples from a population with $\mu = 10$ seconds. Figure 2.10 shows the output from the **Theory-Based Inference** applet for this data set.



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Example 2.3: Estimating Time Conclusions

Formulate Conclusions.

- Based on our small *p*-value, we can conclude that people don't accurately estimate the length of a 10-second song snippet and in fact they overestimate it.
- To what larger population can we make our inference?

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Example 2.3: Summary

- When we test a single quantitative variable, our hypotheses have the following form:
  - $\mathbf{H_0}$: $\mu =$ some number
  - $\mathbf{H_a}$: $\mu \neq <$ or $>$ some number
- We will get our data (or mean, sample size, and SD for our data) and use the Theory-Based Inference applet to determine the *p*-value.
- The *p*-value we get with this test has the same general meaning as those from a test for a single proportion.

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Example 2.3: Decision Errors (1 of 3)

- When we make a conclusion based on a test of significance, hopefully we made the correct decision.
- However, perhaps we didn't.
- In other words, we made a decision that is in error.
- We usually don't know if we made a decision error, but let's look closer at the two types of decision errors we could have made.

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Example 2.3: Decision Errors (2 of 3)

- We rejected the null hypothesis in this study, does that mean that we've *proven* that the population mean is not equal to 10 seconds?
- If the null was true and $\mu = 10$ seconds, (and we assume we had a random sample), we still could have been very unlikely and obtained a sample with too many over-estimators.
- This means we would have found strong evidence that people can't estimate the song snippet correctly, on average, when in fact they can.
- This is a decision error we could have made.

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Example 2.3: Decision Errors (3 of 3)

- What about the opposite case?
- What if we tested H_0: $\mu = 13$ seconds (and with our statistic of 13.71 seconds) our *p*-value had been large so that we did not reject the null hypothesis?
- Would we have *proven* students, on average, estimate the song snippet as being 13 seconds long?
- We would not have strong evidence against the null, but suppose the null was not true.
- This is another type of decision error that we could have made.

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Cool-Down

- *Poll the class for how many hours of sleep they get.*
- A representative of each row should report individual data points for their entire row.
- We'll plug this data into the applet next time to explore.

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